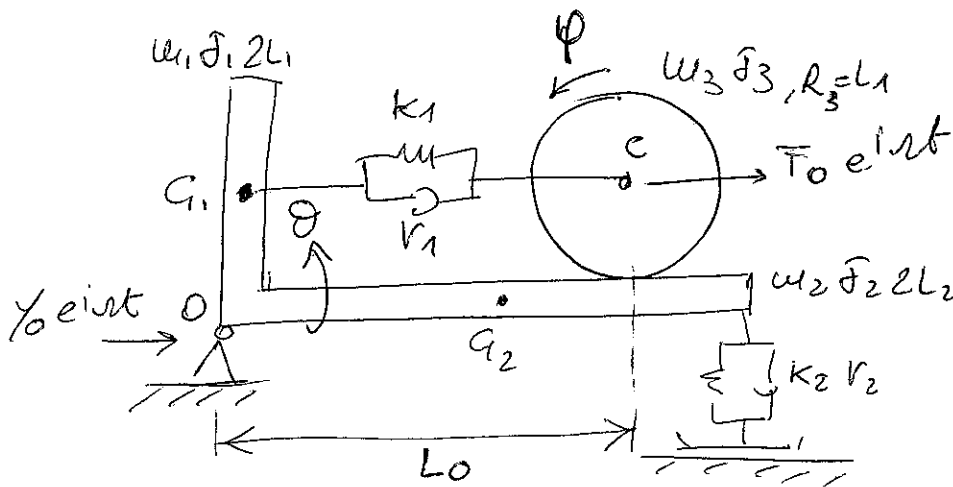


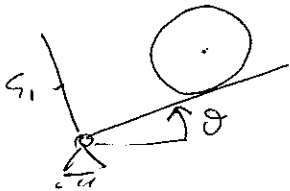


$$\underline{x} = \begin{Bmatrix} \theta \\ \phi \\ y \end{Bmatrix}$$

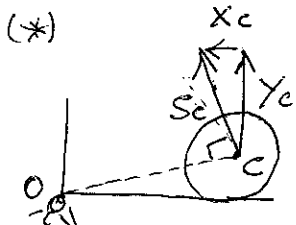
ϕ = rotazione relativa rispetto a θ

$$E_c = \frac{1}{2} m_1 \dot{x}_{G_1}^2 + \frac{1}{2} (J_1 + J_2) \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G_2}^2 + \frac{1}{2} m_2 \dot{y}_{G_2}^2 + \frac{1}{2} m_3 \dot{x}_C^2 + \frac{1}{2} m_3 \dot{y}_C^2 + \frac{1}{2} J_3 \omega_3^2$$

$\theta \neq 0; \phi = y = 0$

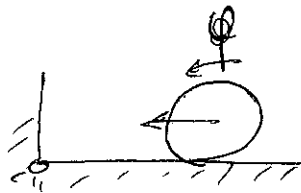


$$\begin{aligned} x_{G_1} &= -L_1 \theta \\ x_{G_2} &= 0 \\ y_{G_2} &= L_2 \theta \\ x_C &= -R_3 \theta \quad (*) \\ y_C &= L_0 \theta \quad (*) \\ \omega_3 &= \theta \end{aligned}$$



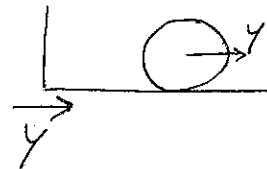
$$\begin{aligned} \Delta L_1 &= 0 \\ \Delta L_2 &= 2L_2 \theta \end{aligned}$$

$\theta = 0; \phi \neq 0; y = 0$



$$\begin{aligned} x_{G_1} &= 0 \\ x_{G_2} &= 0 \\ y_{G_2} &= 0 \\ x_C &= -R_3 \phi \\ y_C &= 0 \\ \omega_3 &= \phi \\ \Delta L_1 &= -R_3 \phi \\ \Delta L_2 &= 0 \end{aligned}$$

$\theta = \phi = 0; y \neq 0$



$$\begin{aligned} x_{G_1} &= y \\ x_{G_2} &= y \\ y_{G_2} &= 0 \\ x_C &= y \\ y_C &= 0 \\ \omega_3 &= 0 \\ \Delta L_1 &= 0 \\ \Delta L_2 &= 0 \end{aligned}$$

$$\begin{cases} x_{s1} = y - L_1 \theta \\ x_{s2} = y \\ y_{s2} = L_2 \theta \\ x_c = -R\theta - R\phi + y \\ y_c = L_0 \theta \\ \omega_3 = \theta + \phi \end{cases}$$

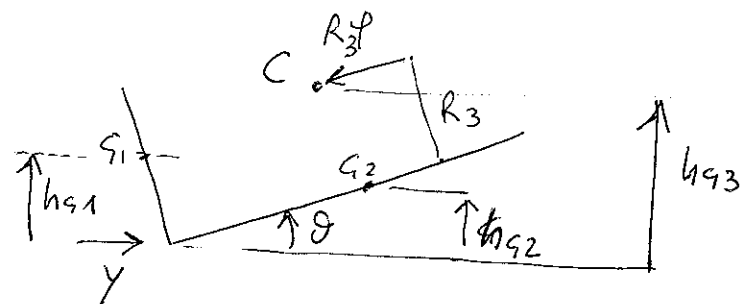
$$[u_F] = \text{diag}(\omega_1 (\dot{\theta}_1 + \dot{\theta}_2) \omega_2 \omega_2 \omega_3 \omega_3 \dot{\theta}_3)$$

$$[\Lambda_u] = \begin{bmatrix} -L_1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ L_2 & 0 & 0 \\ -R_3 & -R_3 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$V_{EL} = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 \quad [k_F] = \text{diag}(k_1 \ k_2)$$

$$\begin{cases} \Delta L_1 = -R_3 \phi \\ \Delta L_2 = 2L_2 \theta \end{cases} \quad [\Lambda_K] = \begin{bmatrix} 0 & -R_3 & 0 \\ 2L_2 & 0 & 0 \end{bmatrix}$$

$$V_s = \omega_1 g h_{s1} + \omega_2 g h_{s2} + \omega_3 g h_{s3}$$



$$h_{s1} = L_1 \cos \theta$$

$$h_{s2} = L_2 \sin \theta$$

$$h_{s3} = L_0 \sin \theta + R_3 \cos \theta - R_3 \phi \sin \theta$$

$$V_s = \omega_1 g L_1 \cos \theta + \omega_2 g L_2 \sin \theta + \omega_3 g (L_0 \sin \theta + R_3 \cos \theta - R_3 \phi \sin \theta)$$

$$\begin{cases} \frac{\partial V_s}{\partial \theta} = -m_1 g L_1 \sin \theta + m_2 g L_2 \cos \theta + m_3 g (L_0 \cos \theta - R_3 \sin \theta - R_3 \phi \cos \theta) \\ \frac{\partial V_s}{\partial \phi} = -m_3 g R_3 \sin \theta \\ \frac{\partial V_s}{\partial \psi} = 0 \end{cases}$$

$$\begin{cases} \frac{\partial^2 V_s}{\partial \theta^2} = -m_1 g L_1 \cos \theta - m_2 g L_2 \sin \theta + m_3 g (-L_0 \sin \theta - R_3 \cos \theta + R_3 \phi \sin \theta) \\ \frac{\partial^2 V_s}{\partial \phi^2} = 0 \\ \frac{\partial^2 V_s}{\partial \psi^2} = \frac{\partial^2 V_s}{\partial \theta \partial \psi} = \frac{\partial^2 V_s}{\partial \phi \partial \psi} = 0 \\ \frac{\partial^2 V_s}{\partial \theta \partial \phi} = -m_3 g R_3 \cos \theta \end{cases}$$

$$[k_s] = \begin{bmatrix} -m_1 g L_1 - m_3 g R_3 & -m_3 g R_3 & 0 \\ -m_3 g R_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$[k_{II}] = 0$ x de ΔL_1 depende linearmente de ϕ
e ΔL_2 depende de $\sin \theta$

$$D = \frac{1}{2} r_1 \dot{\Delta l}_1^2 + \frac{1}{2} \dot{\Delta l}_2^2$$

$$[r_F] = \text{diag}(r_1, r_2)$$

molle in // agli smorzatori $\rightarrow [A_1] = [A_2]$

$$\delta L = F_0 \delta x_c + F_y \delta y = F_0 (\delta y - R_3 \delta \varphi - R_3 \delta \theta) + F_y \delta y$$

$$\underline{Q}_L = \begin{Bmatrix} -R_3 \\ -R_3 \end{Bmatrix} F_0 e^{i\omega t} \quad \underline{Q}_V = \{ F_0 + F_y \}$$

domanda 3) rotazione spazzante = ϑ
rotazione assoluta disco = $\vartheta + \varphi$

domanda 4) spost. orig. C : x_c

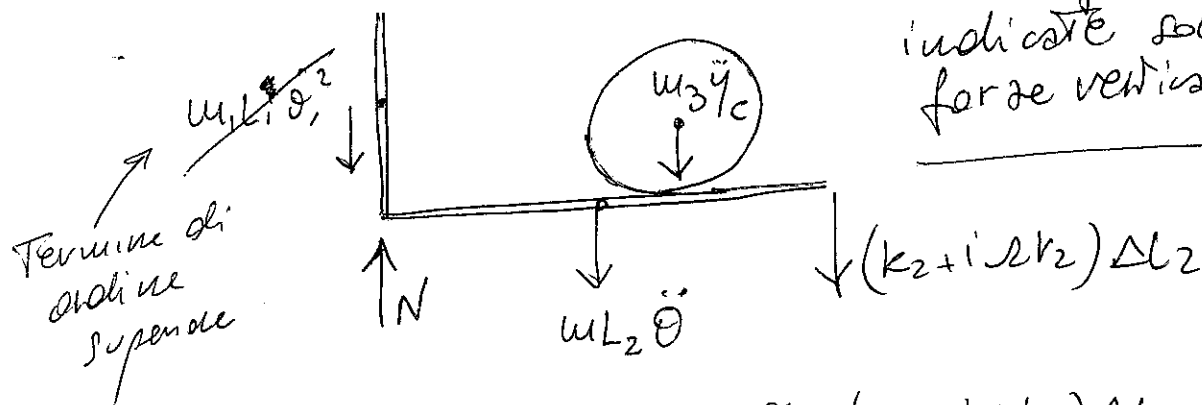
Forze trasmesse da k_2, r_2 : $F_2 = (k_2 + i\omega r_2) \Delta l_2$

$$\Delta l_2 = 2L_2 \theta$$

domanda 5) rotazione relativa disco = φ

domanda 6)

Nel disegno sono indicate solo le forze verticali



$$\sum F_v = 0 : N = w_2 L \ddot{\theta} + w_3 L \ddot{\theta} + (k_2 + i\omega r_2) \Delta l_2$$

$$y_c = L \theta \rightarrow \ddot{y}_c = L \ddot{\theta}$$